

NexusMUD Maker

ChronicleGroundTruth API

Integration Spec

Companion Engineering Reference — Chronicle Ingestion Pipeline

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1. Overview

1.1 Purpose

The GroundTruth API is the authoritative world-state data source for NexusMUD. The Maker Chronicle ingestion pipeline queries GroundTruth to fulfil the following core functions:

- Validate AscendantEvent payloads against live world-state
- Enrich Chronicle entries with authoritative location, entity, and item data
- Resolve `makerId` and `sigilAsset` references
- Confirm area access tiers and ecology metadata

This document is the primary engineering reference for all Chronicle-GroundTruth integration work. It is implementation-ready and intended for direct use by backend engineers during sprint execution.

1.2 Integration Scope

In Scope

- Read-only GroundTruth queries during Chronicle entry creation and curation
- Webhook / event subscription for AscendantEvent delivery
- Maker profile resolution (epithet, sigil, attribution mode)
- Area, item, entity, lore, and system record fetches

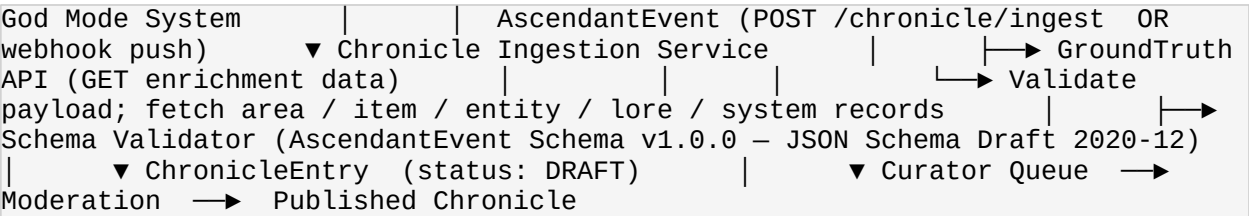
Out of Scope

- Writing back to GroundTruth from the Chronicle pipeline (Chronicle is **read-only** against GroundTruth)

- GroundTruth admin or world-state mutation endpoints
- Player-facing GroundTruth queries (Chronicle API is the player interface — GroundTruth is never exposed directly)

1.3 Architecture Position

The Chronicle ingestion pipeline occupies the enrichment and curation layer between the God Mode system and the player-visible Chronicle. The authoritative data flow is as follows:



NOTE — GroundTruth Authoritativeness

GroundTruth values are always authoritative. When a conflict exists between an AscendantEvent payload field and the corresponding GroundTruth record, the GroundTruth value **overwrites** the payload value in the ChronicleEntry draft. All conflicts are logged to CurationRecord for curator review. See Section 5.4.

2. Authentication

2.1 Auth Method

The GroundTruth API uses JWT Bearer authentication. The Chronicle Ingestion Service authenticates as a service account with the scope `chronicle:read`. All credentials are sourced from the secrets manager at runtime — never from static config files or source code.

Field	Value
Auth Type	Bearer JWT
Token Endpoint	POST /auth/token

Field	Value
Grant Type	client_credentials
Required Scope	chronicle:read
Token TTL	3600 seconds (1 hour)
Refresh Strategy	Proactive re-request when remaining TTL < 300 seconds (configurable via GT_TOKEN_TTL_BUFFER_SECS)
Secret Storage	Environment variable GT_CLIENT_SECRET — injected from Key Vault at deploy time. Never hardcoded.

2.2 Token Request Example

Request (curl):

```
curl -X POST https://groundtruth.nexusmud.internal/auth/token \
  -H "Content-Type: application/x-www-form-urlencoded" \
  -d "grant_type=client_credentials\ &client_id=${GT_CLIENT_ID}\ &client_secret=${GT_CLIENT_SECRET}\ &scope=chronicle:read"
```

Response (200 OK):

```
{  "access_token": "eyJhbGciOiJIUzI1NiIs... ",  "token_type": "Bearer",
  "expires_in": 3600,  "scope": "chronicle:read" }
```

Store access_token in-memory. Never write to disk. Include as Authorization:

Bearer <token> on every subsequent GroundTruth request.

2.3 Auth Error Handling

HTTP Status	Error Code	Required Action
401	invalid_token	Re-authenticate immediately. Log incident to security event stream. Do not cache the failed token.
403	insufficient_scope	Alert ops immediately. Do not retry — this is a scope configuration error requiring manual intervention.
429	rate_limited	Apply backoff per Section 5.2. Do not re-authenticate — the token is valid.

HTTP Status	Error Code	Required Action
503	auth_unavailable	Queue the inbound event in the durable retry queue. Retry auth after 60 seconds.

3. GroundTruth API Endpoints Used by Chronicle

All endpoints below are read-only (GET or POST batch-read). The Chronicle Ingestion Service holds the `chronicle:read` scope only. Mutation endpoints are not accessible to this service account and must not be called.

3.1 Maker Profile Resolution

GET `/makers/{makerId}`

Description: Fetch the authoritative maker profile, including epithet, sigil, and attribution preferences. Called on every AscendantEvent ingestion to resolve maker display identity before the ChronicleEntry draft is created.

Parameter	Type	Location	Description
makerId	string	Path	Format: mk-NNNN-slug (e.g., mk-0041-ares)

Response Fields: `makerId`, `epithet`, `sigilAsset`, `attributionMode`, `domainPrimary`, `domainSecondary[]`, `accountStatus`

Business Rule: If `accountStatus` is not "active", set `moderationHooks.autoFlag = true`, `flagReason = "Maker account not active"`. Do not auto-publish the entry.

```
{  "makerId": "mk-0041-ares",    "epithet": "The Pale Shaper",
  "sigilAsset": "sigil_arc_pale_01",  "attributionMode": "pseudonymous",
  "domainPrimary": "Genesis",    "domainSecondary": ["Weave"],
  "accountStatus": "active" }
```

3.2 Area Record Fetch

GET `/world/areas/{areaId}`

Description: Fetch authoritative area data for `AREA_CREATED` / `AREA_MODIFIED` events. Enriches Genesis domain Chronicle entries with verified area name, tier, and ecology description.

Parameter	Type	Location	Description
areaId	string	Path	Unique area identifier
include	string	Query	ecology, coordinates, linkedEntities, boundaryLayer

Response Fields: areaId, name, accessTier, coordinates, ecology, boundaryLayerStatus, linkedEntities[], createdAt, lastModifiedAt

Business Rule: GroundTruth name and accessTier overwrite payload values on conflict. If linkedEntities is non-empty, execute a batch fetch (Section 3.7) for all referenced entity IDs.

```
{  "areaId": "area-stormfall-001",  "name": "Stormfall Reach",  "accessTier": 3,  "coordinates": { "x": 44.7, "y": -12.3, "z": 0 },  "ecology": "Storm-touched fauna, perpetual electrical storms, basalt spire formations.",  "boundaryLayerStatus": "STABLE",  "linkedEntities": ["entity-hungering-veil-01"],  "createdAt": "2026-06-10T06:14:00Z",  "lastModifiedAt": "2026-06-10T06:14:00Z" }
```

3.3 Item / Forge Record Fetch

GET /world/items/{itemId}

Description: Fetch authoritative item record for `POWER_FORGE` / `ITEM_CREATED` events. Validates Forge domain payload; resolves resonance rating and replication status for blurb generation and suggested artifact list.

Parameter	Type	Location	Description
itemId	string	Path	Unique item identifier
include	string	Query	materials, resonance, replication

Response Fields: itemId, name, itemClass, resonanceRating, isUnique, replicationStatus, forgeLocation, materials[], custodyLocation, createdAt

```
{  "itemId": "item-sundering-blade-42",  "name": "The Sundering Blade",
  "itemClass": "weapon",  "resonanceRating": 8.7,  "isUnique": true,
  "replicationStatus": "UNREPLICABLE",  "forgeLocation": "The Deep Crucible,
  Ashvein Mountains",  "materials": [    { "materialId": "mat-veldrite-
  shard", "materialName": "Veldrite Shard", "quantity": 7 }  ],
  "custodyLocation": "The Deep Crucible Vault",  "createdAt": "2026-06-
  09T22:47:00Z" }
```

3.4 Entity / Boss Record Fetch

GET /world/entities/{entityId}

Description: Fetch authoritative entity record for BOSS_SPAWN / ENTITY_CREATED events. Enriches Genesis / Weave domain entries; tier and loreClassification drive moderationHooks routing.

Parameter	Type	Location	Description
entityId	string	Path	Unique entity identifier
include	string	Query	drops, respawn, loreClassification

Response Fields: entityId, name, entityType, tier, spawnLocation, respawnCycleSecs, associatedDrops[], loreClassification, createdAt

Business Rule: If tier >= 5 OR loreClassification == "LEGEND", set moderationHooks.autoFlag = true, priorityLevel = "urgent".

```
{  "entityId": "entity-hungering-veil-01",  "name": "The Hungering Veil",
  "entityType": "boss",  "tier": 5,  "spawnLocation": "Stormfall Reach –
  Inner Veil Chamber",  "respawnCycleSecs": 86400,  "associatedDrops":
  ["item-veil-shard-rare", "item-encounter-charter-01"],
  "loreClassification": "LEGEND",  "createdAt": "2026-06-05T03:30:00Z" }
```

3.5 Lore Record Fetch

GET /world/lore/{loreId}

Description: Fetch authoritative lore record for LORE_ENTRY / LORE_REVISED events. Chronicle domain entries pull canonStatus and sourceweight directly from GroundTruth; blurb templates use these values.

Parameter	Type	Location	Description
loreId	string	Path	Unique lore identifier

Parameter	Type	Location	Description
include	string	Query	sources, disputes, canon

Response Fields: loreId, name, canonStatus, sourceWeight, primarySources[], disputedClaims[], accessLevel, lastRevisedAt

```
{  "loreId": "lore-ashen-compact-001",  "name": "The Ashen Compact",  "canonStatus": "CANON",  "sourceWeight": 0.85,  "primarySources": ["src-ashen-archives-vol3", "src-elder-testimony-12"],  "disputedClaims": [],  "accessLevel": "public",  "lastRevisedAt": "2026-06-07T11:00:00Z" }
```

3.6 System / Calibration Record Fetch

GET /world/systems/{systemId}

Description: Fetch authoritative Clockwork system record for CALIBRATION events. Clockwork domain blurb templates pull tolerance band, post-calibration metrics, and failsafe status from GroundTruth for precision rendering.

Parameter	Type	Location	Description
systemId	string	Path	Unique system identifier
include	string	Query	metrics, flags, failsafe

Response Fields: systemId, name, cycleId, toleranceBand, postCalibrationMetrics{}, openFlags[], failsafeStatus, lastCalibratedAt

Business Rule: If failsafeStatus == "TRIGGERED", set moderationHooks.autoFlag = true, priorityLevel = "urgent".

```
{  "systemId": "sys-gravetide-engine-001",  "name": "Gravetide Engine",  "cycleId": 44,  "toleranceBand": { "min": 0.0, "max": 0.005, "unit": "pct_drift" },  "postCalibrationMetrics": { "drift_pct": 0.003, "cycle_integrity": 99.997 },  "openFlags": ["flag-secondary-resonance", "flag-failsafe-F3"],  "failsafeStatus": "ARMED",  "lastCalibratedAt": "2026-06-06T08:22:00Z" }
```

3.7 Bulk Enrichment Batch Endpoint

POST /world/batch

Description: Fetch multiple record types in a single request. Use this endpoint to reduce round-trips during complex event ingestion — for example, an AREA_CREATED

event that also references spawned entities and created items. Preferred over sequential individual fetches.

Constraint	Value
Max records per request	20
Supported types	area, item, entity, lore, system, maker
Auth required	Yes — same Bearer JWT as individual endpoints
Partial failure behavior	Response returns error objects per failed ID; successfully resolved records are still returned

Request Body:

```
{  "records": [    { "type": "area",   "id": "area-stormfall-001" },    { "type": "entity", "id": "entity-hungering-veil-01" },    { "type": "maker",  "id": "mk-0041-ares" }  ] }
```

Response (200 OK):

```
{  "area:area-stormfall-001": {    "areaId": "area-stormfall-001",    "name": "Stormfall Reach",    "accessTier": 3,    "...": {}  },  "entity:entity-hungering-veil-01": {    "entityId": "entity-hungering-veil-01",    "name": "The Hungering Veil",    "tier": 5,    "...": {}  },  "maker:mk-0041-ares": {    "makerId": "mk-0041-ares",    "epithet": "The Pale Shaper",    "accountStatus": "active",    "...": {}  } }
```

Chronicle Use Case: Called when an `AscendantEvent` payload contains non-empty `linkedEntities` or `associatedDrops` arrays. All referenced IDs are bundled into a single batch request rather than issued as sequential individual GETs.

4. Event Ingestion Flow (Detailed)

4.1 Step-by-Step Ingestion Procedure

1. Chronicle Ingestion Service receives `AscendantEvent` via `POST /chronicle/ingest` (webhook push) or via poll (fallback — see Section 6.1 Mode B).
2. Validate event envelope against `AscendantEvent Schema v1.0.0` (JSON Schema Draft 2020-12). On validation failure: store event record as `FAILED`,

write validation errors to `ingestionMeta.validationErrors`, alert ops. Do not continue.

3. Deduplicate: check ingestion log for `eventId`. If already present: store as `DUPLICATE`, return `200 OK` (idempotent). Do not re-process.
4. Resolve `makerId` via `GET /makers/{makerId}`. If `accountStatus != "active"`: set `moderationHooks.autoFlag = true`, `flagReason = "Maker account not active"`. Continue — do not abort.
5. Determine enrichment targets from `eventType` using the Event-Type Mapping Table (Section 4.2).
6. Execute batch enrichment via `POST /world/batch` for all referenced IDs. Observe batch size limit of 20; issue multiple batch requests if necessary.
7. Merge `GroundTruth` data into `ChronicleEntry` draft fields. `GroundTruth` values are authoritative and **overwrite** event payload values on conflict. Log all conflicts per Section 5.4.
8. Evaluate `ModerationHook` rules (Section 4.3). Set `autoFlag` and `priorityLevel` accordingly.
9. Select blurb template based on domain and tone (Section 4.4). Pre-populate bracket fields from `GroundTruth` data where direct mapping exists (Section 4.5). Leave unresolved brackets for curator completion.
10. Create `ChronicleEntry` with `status = DRAFT` and write to Chronicle database.
11. Route to curator queue (if `autoFlag = false`) or moderation queue (if `autoFlag = true`).
12. Update `ingestionMeta.ingestionStatus = INGESTED`; set `chronicleEntryId`.
13. Return `201 Created` with `chronicleEntryId` and `status` in response body.

4.2 Event-Type to Enrichment Target Mapping

Event Type	Domain	GroundTruth Endpoints Called	Template Family	Auto-Flag Rule
AREA_CREATED	Genesis	/makers/{id}, /world/areas/{id}, /world/batch (linkedEntities)	genesis	Never auto (unless maker inactive)
AREA_MODIFIED	Genesis	/makers/{id}, /world/areas/{id}	genesis	Never auto
AREA_DELETED	Genesis	/makers/{id}, /world/areas/{id}	genesis	Always auto-flag (destructive)
POWER_FORGE	Forge	/makers/{id}, /world/items/{id}	forge	If isUnique = true → elevated
ITEM_CREATED	Forge	/makers/{id}, /world/items/{id}	forge	Never auto
ITEM_MODIFIED	Forge	/makers/{id}, /world/items/{id}	forge	Never auto
ITEM_DELETED	Forge	/makers/{id}, /world/items/{id}	forge	Always auto-flag (destructive)
BOSS_SPAWN	Genesis	/makers/{id}, /world/entities/{id}, /world/batch (drops)	genesis	If tier >= 5 → urgent
ENTITY_CREATED	Genesis / Weave	/makers/{id}, /world/entities/{id}	genesis	If tier >= 4 → elevated
ENTITY_MODIFIED	Genesis / Weave	/makers/{id}, /world/entities/{id}	genesis	Never auto
ENTITY_DELETED	Genesis / Weave	/makers/{id}, /world/entities/{id}	genesis	Always auto-flag (destructive)
LORE_ENTRY	Chronicle	/makers/{id}, /world/lore/{id}	chronicle	If canonStatus = DISPUTED → elevated
LORE_REVISIED	Chronicle	/makers/{id},	chronicle	Always

Event Type	Domain	GroundTruth Endpoints Called	Template Family	Auto-Flag Rule
		/world/lore/{id}		elevated
LORE_REDACTED	Chronicle	/makers/{id}, /world/lore/{id}	chronicle	Always urgent
CALIBRATION	Clockwork	/makers/{id}, /world/systems/{id}	clockwork	If failsafeStatus = TRIGGERED → urgent
SYSTEM_MODIFIED	Clockwork	/makers/{id}, /world/systems/{id}	clockwork	Never auto
SYSTEM_CREATED	Clockwork	/makers/{id}, /world/systems/{id}	clockwork	Never auto
FACTION_CREATED	Chronicle	/makers/{id}	chronicle	Never auto
QUEST_CREATED	Genesis	/makers/{id}	genesis	Never auto
NPC_CREATED	Genesis	/makers/{id}, /world/entities/{id}	genesis	Never auto

4.3 ModerationHook Evaluation Rules

Rules are evaluated in order for each ingested event. `priorityLevel` resolution: **urgent > elevated > normal** (highest wins). `autoFlag` is OR'd — any rule producing `true` sets the final value to `true`.

1. If `maker.accountStatus != "active"` → `autoFlag = true`, `priorityLevel = "urgent"`, `flagReason = "Maker account not active"`
2. If `eventType` contains "DELETED" → `autoFlag = true`, `priorityLevel = "elevated"`, `flagReason = "Destructive event type"`
3. If `entity.tier >= 5` OR `entity.loreClassification == "LEGEND"` → `autoFlag = true`, `priorityLevel = "urgent"`

4. If `entity.tier == 4` → `priorityLevel = "elevated"` (no `autoFlag`)
5. If `item.isUnique == true` → `priorityLevel = "elevated"`
6. If `lore.canonStatus == "DISPUTED"` → `priorityLevel = "elevated"`
7. If `eventType == "LORE_REDACTED"` → `autoFlag = true`, `priorityLevel = "urgent"`
8. If `system.failSAFEStatus == "TRIGGERED"` → `autoFlag = true`, `priorityLevel = "urgent"`
9. If `eventType == "LORE_REVISED"` → `priorityLevel = "elevated"` (minimum; may be overridden to urgent by other rules)
10. Default: `autoFlag = false`, `priorityLevel = "normal"`

4.4 Blurb Template Routing

The ingestion pipeline selects the initial blurb template based on domain and default tone. Curators may override tone selection in the Curation Editor at any point before submission for approval.

Domain	Default Tone	Template IDs Available	Override Mechanism
Weave	cryptic	<code>tmpl-weave-short</code> , <code>tmpl-weave-medium</code> , <code>tmpl-weave-long</code>	Curator tone selector in Curation Editor
Forge	heroic	<code>tmpl-forge-short</code> , <code>tmpl-forge-medium</code> , <code>tmpl-forge-long</code>	Curator override
Genesis	epic	<code>tmpl-genesis-short</code> , <code>tmpl-genesis-medium</code> , <code>tmpl-genesis-long</code>	Curator override
Chronicle	scholarly	<code>tmpl-chronicle-short</code> , <code>tmpl-chronicle-medium</code> , <code>tmpl-chronicle-long</code>	Curator override

Domain	Default Tone	Template IDs Available	Override Mechanism
Clockwork	mechanical	tmpl-clockwork-short, tmpl-clockwork-medium, tmpl-clockwork-long	Curator override

The ingestion pipeline pre-populates bracket fields from GroundTruth data where a direct mapping exists (Section 4.5). Remaining unfilled brackets are left for curator completion at the Curation Editor step.

4.5 GroundTruth Field-to-Template Bracket Mapping

Template Bracket	GroundTruth Source Field	Fallback if Null
[MAKER_EPITHET]	maker.epithet	"An Unknown Maker"
[LOCATION_NAME]	area.name	payload.areaName
[ACCESS_RATING]	area.accessTier — formatted as "Tier N"	"Unrated"
[ECOLOGY_OR_ARCHITECTURE]	area.ecology	"[ecology pending survey]"
[ITEM_NAME]	item.name	payload.itemName
[RESONANCE_RATING]	item.resonanceRating	"[unrated]"
[REPLICATION_STATUS]	item.replicationStatus	"unknown"
[ENTITY_NAME]	entity.name	payload.entityName
[TIER]	entity.tier	"[untiered]"
[CANON_STATUS]	lore.canonStatus	"PENDING"
[SOURCE_WEIGHT]	lore.sourceWeight — formatted to 2 decimal places	"[weight pending]"
[FAILSAFE_STATUS]	system.failsafeStatus	"[status unknown]"
[CYCLE_ID]	system.cycleId	"[cycle unknown]"

5. Error Handling & Resilience

5.1 GroundTruth Unavailability Strategy

The Chronicle Ingestion Service must not block or discard events during GroundTruth outages. The following graceful degradation procedure applies when the GroundTruth API is unreachable:

14. On first connection failure, immediately store the inbound event in the durable retry queue (Redis Streams or equivalent persistent queue). Do not drop the event.
15. Retry enrichment with exponential backoff: attempts at 5s, 15s, 45s, 135s, 405s intervals.
16. A ChronicleEntry is **not** created until enrichment succeeds or the 2-hour enrichment timeout elapses (configurable via `GT_ENRICHMENT_TIMEOUT_SECS`).
17. On 2-hour timeout expiry: create ChronicleEntry with `status = DRAFT`; populate fields from event payload only (no GroundTruth data). Set `ingestionMeta.ingestionStatus = "PARTIAL"`. Append curator warning: *"⚠ GroundTruth enrichment unavailable — manual field verification required before publication."*
18. Alert ops and Curation lead when any entry enters PARTIAL status.

WARNING — PARTIAL Entries

PARTIAL ChronicleEntries must never be auto-published. They must be routed to the moderation queue with `autoFlag = true` regardless of the normal ModerationHook evaluation result. This is enforced at the ChronicleEntry write step.

5.2 Rate Limiting

The GroundTruth API enforces a limit of **300 requests per minute** per service account (configurable reference value: `GT_RATE_LIMIT_RPM`).

Pipeline strategies to stay within limit:

- The batch endpoint (POST /world/batch) is the preferred fetch mechanism — reduces N individual GETs to a single request for complex events.
- In-process in-memory cache with a 60-second TTL (GT_CACHE_TTL_SECS) for maker profiles and area / item / entity records. Immutable during a single ingestion transaction — do not re-fetch mid-flow.
- On receipt of HTTP 429: read the Retry-After response header. Wait for the specified duration (default 30 seconds if header is absent), then retry. Maximum 3 retries (GT_RETRY_MAX) before transferring the event to the durable retry queue.

5.3 Partial Enrichment

If one or more sub-requests within a batch call fail (e.g., a single entity ID returns 404 or 503) but others succeed, the pipeline proceeds as follows:

19. Log failed record IDs and error codes to `ingestionMeta.validationErrors[]`.
20. Continue with partial enrichment — populate template bracket fields for all successfully resolved records.
21. Leave unresolved bracket fields blank; they will be flagged in the Curation Editor for manual completion.
22. Set `ChronicleEntry.status = DRAFT`. Append curator warning: "*⚠ Partial enrichment: the following IDs could not be resolved — [list IDs]. Manual verification required for affected fields.*"
23. Route normally (partial enrichment alone does not trigger `autoFlag`), but the curator warning will prevent publication without acknowledgement.

5.4 Conflict Resolution

When a GroundTruth record value disagrees with the corresponding field in the AscendantEvent payload, the following authority rules apply:

Authority	Authoritative Fields
GroundTruth (always wins)	name, accessTier, resonanceRating, tier, canonStatus, sourceWeight,

Authority	Authoritative Fields
	failsafeStatus
AscendantEvent Payload (authoritative)	timestamp, eventType, attributionMode (maker's own declared preference), tags (pre-assigned by God Mode system)

All conflicts are logged to CurationRecord with the field key "gt_conflict", storing both the original payload value and the GroundTruth override value. Curators see a conflict diff in the Curation Editor Diff View. The Curation Editor Diff View is a curator-only interface — see Section 7 item 8.

6. Webhook / Event Subscription

6.1 AscendantEvent Delivery Mechanism

Two delivery modes are supported. Mode A (Push) is the production default. Mode B (Poll) is a circuit-breaker fallback only.

Mode A — Push (Webhook) — Primary

- The God Mode system POSTs an AscendantEvent JSON payload to `POST /chronicle/ingest` on the Chronicle Ingestion Service.
- Authentication: HMAC-SHA256 signature in the `X-NexusMUD-Signature` header (format: `sha256=<hex_digest>`). Secret is shared between God Mode and the Chronicle service via Key Vault (`GT_WEBHOOK_SECRET`).
- Chronicle Ingestion Service verifies the HMAC signature before processing (see Section 6.2). Payloads failing signature verification are rejected with `401 Unauthorized` and logged to the security event stream.
- Response: `202 Accepted` immediately. Processing is fully asynchronous — the `202` only acknowledges receipt and signature validity.
- Retry policy (God Mode side): 3 retries with exponential backoff on `4xx` / `5xx` responses. After 3 failures, God Mode logs the event as undelivered and alerts ops.

Mode B — Poll (Fallback) — Circuit Breaker Only

- Chronicle Ingestion Service polls: `GET /events?since={lastProcessedTimestamp}&domain=all&limit=50`
- Poll interval: 60 seconds.
- Pagination: use `nextCursor` from response body for subsequent pages. Stop pagination when `nextCursor` is `null`.
- Mode B is activated only when the webhook delivery circuit breaker enters open state. Resume Mode A as soon as webhook delivery is confirmed restored.

6.2 Webhook Signature Verification

All inbound webhook payloads must be verified before processing. Use constant-time comparison to prevent timing attacks.

```
import hmac, hashlib def verify_signature(payload_bytes: bytes, header_sig: str, secret: str) -> bool:     expected = hmac.new(secret.encode(), payload_bytes, hashlib.sha256).hexdigest()     received = header_sig.removeprefix("sha256=")     return hmac.compare_digest(expected, received) # Usage in request handler: raw_body = request.get_raw_body() # bytes – do NOT decode before verification sig_header = request.headers.get("X-NexusMUD-Signature", "") if not verify_signature(raw_body, sig_header, GT_WEBHOOK_SECRET): log_security_event("webhook_sig_failure", sig_header)     return Response(status=401)
```

IMPORTANT

Always compute the HMAC over the **raw bytes** of the request body, before any JSON parsing or decoding. Computing over a decoded or re-serialized payload will produce signature mismatches for payloads with non-ASCII characters or whitespace differences.

6.3 Idempotency

Every `AscendantEvent` carries a unique `eventId` (format: UUID v4) assigned by the God Mode system at emission time. The Chronicle Ingestion Service maintains a persistent ingestion log keyed on `eventId` in the Chronicle database.

- On receipt: check ingestion log for `eventId` before any processing begins.

- If `eventId` is already present with `ingestionStatus = INGESTED`: return `200 OK` with the existing `chronicleEntryId`. No re-processing occurs.
- If `eventId` is present with `ingestionStatus = FAILED`: treat as a new delivery attempt — proceed with normal ingestion. This allows ops to trigger re-delivery of previously failed events.
- The ingestion log entry is written atomically with the `ChronicleEntry` creation to prevent partial-write race conditions.

7. Security Considerations

24. **Transport security:** All GroundTruth API calls use HTTPS with TLS 1.2 minimum (TLS 1.3 preferred). No plaintext HTTP is permitted in any environment, including local dev. The local dev mock server (Section 10.A) uses HTTP — this exception applies only to the mock and must not propagate to staging or production.
25. **Secrets management:** `GT_CLIENT_SECRET` and `GT_WEBHOOK_SECRET` are stored in the secrets manager (e.g., Azure Key Vault). They are injected at deploy time as environment variables. They must never appear in configuration files, source code, Docker images, or environment variable dumps in logs.
26. **Minimum scope:** The Chronicle Ingestion Service service account requests only the `chronicle:read` scope. No write scopes are requested or granted. If a `403 insufficient_scope` is received, the correct resolution is a configuration fix — not a scope escalation request.
27. **Webhook HMAC rotation:** `GT_WEBHOOK_SECRET` is rotated quarterly. Rotation procedure: update the secret in Azure Key Vault; simultaneously re-deploy both God Mode and Chronicle Ingestion Service during a low-traffic maintenance window. The window must be scheduled and communicated to the Curation lead and ops at least 48 hours in advance.

28. **GroundTruth response treatment:** API responses are treated strictly as structured data. No `eval()`, `exec()`, or dynamic code execution of response content is permitted anywhere in the ingestion pipeline.
29. **Anonymous maker log redaction:** If a maker's `attributionMode == "anonymous"`, the `makerId` value is redacted from all ingestion log entries (replaced with `"[REDACTED]"`). The non-redacted value is stored only in the `ChronicleEntry`'s secured fields, accessible to curators with appropriate permissions.
30. **Security event stream:** Rate limit failures (429), auth failures (401, 403), and webhook signature verification failures are logged to the dedicated security event stream (separate from the application event log). These events are monitored for anomaly detection.
31. **Curator Diff View access control:** The Curation Editor Diff View (which exposes GroundTruth vs. payload conflict data) is a curator-only interface. Auth middleware enforcing the `curator` role must be applied at the route level. This view must not be exposed to standard player-facing API routes.

8. Monitoring & Alerting

All metrics below are emitted by the Chronicle Ingestion Service and ingested into the NexusMUD operations monitoring platform. Alert thresholds and targets are advisory defaults — calibrate in staging before production cutover.

Metric	Type	Alert Threshold	Alert Target	Description
gt_api_latency_p99	Gauge	> 2000 ms	Ops channel	GroundTruth API 99th percentile response latency. Sustained high latency will delay Chronicle

Metric	Type	Alert Threshold	Alert Target	Description
				entry creation and increase retry queue depth.
gt_api_error_rate	Rate	> 5% over 5 min	Ops + Prime Architect	GroundTruth error rate spike. May indicate API instability or a configuration regression. Requires immediate investigation.
ingestion_queue_depth	Gauge	> 100 events	Ops channel	Count of events pending in the durable retry queue. Elevated depth indicates GroundTruth unavailability or a processing bottleneck.
ingestion_failure_rate	Rate	> 2% over 10 min	Ops + Prime Architect	Rate of events reaching FAILED status (schema validation, unrecoverable errors). Indicates upstream AscendantEvent schema violations or ingestion bugs.
auth_token_refresh_failures	Counter	Any failure	Security stream	JWT proactive refresh failures. A non-zero value requires immediate

Metric	Type	Alert Threshold	Alert Target	Description
				investigation to prevent service account lockout.
duplicate_event_rate	Rate	> 10% over 1 hr	Ops channel	High duplicate rate may indicate a God Mode retry storm or eventId generation regression. Idempotency handling will suppress duplicate processing but the rate itself warrants investigation.
partial_enrichment_rate	Rate	> 15% over 1 hr	Ops + Curation lead	Elevated rate of ChronicleEntries with PARTIAL enrichment status. Indicates persistent GroundTruth data gaps or batch sub-request failures; increases curator manual workload.
moderation_queue_depth	Gauge	> 50 entries	Curation lead	Count of ChronicleEntry drafts awaiting curator or moderation review. A sustained backlog may

Metric	Type	Alert Threshold	Alert Target	Description
				require temporary curation capacity increase.

9. Environment Configuration

All runtime configuration is supplied via environment variables. Secret variables are injected from Azure Key Vault at deploy time and must never be set in source-controlled configuration files.

Variable	Description	Example Value	Secret?
GT_API_BASE_URL	GroundTruth API base URL	https://groundtruth.nexus.mud.internal	No
GT_CLIENT_ID	Service account client ID	chronicle-ingest-svc	No
GT_CLIENT_SECRET	Service account client secret for JWT token request	<i>[injected from Key Vault]</i>	YES
GT_WEBHOOK_SECRET	HMAC-SHA256 secret for inbound webhook signature verification	<i>[injected from Key Vault]</i>	YES
GT_TOKEN_TTL_BUFFER_SECS	Seconds before JWT expiry to trigger proactive token refresh	300	No
GT_RATE_LIMIT_RPM	Configured GroundTruth rate limit (requests per minute). Used by the pipeline rate limiter; update if GroundTruth team changes the limit.	300	No
GT_BATCH_MAX_SIZE	Maximum records	20	No

Variable	Description	Example Value	Secret?
	per batch enrichment request to /world/batch		
GT_RETRY_MAX	Maximum retry attempts on 429 or transient errors before entering durable retry queue	3	No
GT_CACHE_TTL_SECS	In-process cache TTL (seconds) for enrichment data	60	No
GT_ENRICHMENT_TIMEOUT_SECS	Maximum wait time (seconds) for enrichment before PARTIAL fallback is triggered	7200	No
CHRONICLE_INGEST_ENDPOINT	This service's webhook reception path (relative)	/chronicle/ingest	No
CHRONICLE_DB_URL	Chronicle database connection string	<i>[injected from Key Vault]</i>	YES

10. Appendix

A. GroundTruth API Base URL Environments

Environment	Base URL	Notes
Local Dev	http://localhost:8090	GroundTruth mock server. HTTP only — exception to TLS rule. Do not use mock URL outside local dev.
Staging	https://groundtruth-staging.nexusmud.international	Mirrors production schema. Data is synthetic — seeded from the Staging World-State

Environment	Base URL	Notes
		Fixture Set.
Production	https://groundtruth.nexusmud.internal	Live world-state. All calls are read-only. Rate limits enforced at 300 rpm per service account.

B. Companion Document References

Document	Version	Status	Description
Maker Chronicle Design Package	v0.1	DRAFT	Full UI spec, data model, moderation workflow, and curator interface wireframes
AscendantEvent Schema	v1.0.0	STABLE	JSON Schema (Draft 2020-12) for the AscendantEvent envelope — the canonical validation reference for Step 2 of the ingestion procedure
NexusMUD God Mode API Reference	TBD	PENDING	Upstream event source documentation. Coordinate with God Mode team for access.
GroundTruth API Reference	TBD	PENDING	Full endpoint documentation for all GroundTruth routes. Request from GroundTruth team — not yet publicly distributed.

C. Glossary

Term	Definition
AscendantEvent	A structured JSON event emitted by the God Mode system whenever a maker changes world state. The canonical input to the Chronicle ingestion pipeline.
ChronicleEntry	A curated record of a maker's world-shaping action. Visible to players only after curator review and moderation approval.
GroundTruth	The authoritative NexusMUD world-state data store and API. GroundTruth data always overrides payload data in cases of conflict.
Ingestion Pipeline	The Chronicle Ingestion Service component responsible for receiving, validating, enriching, and routing AscendantEvents into ChronicleEntry drafts.
makerId	Unique identifier for a God Mode maker account. Format: mk-NNNN-slug (e.g., mk-0041-ares).
moderationHook	A metadata block on a ChronicleEntry that controls curator queue routing, auto-flagging, priority level, and flag reasons. Populated during ingestion by the ModerationHook Evaluation Rules (Section 4.3).
Blurb Template	A domain-specific lore text template containing bracket substitution fields (e.g., [MAKER_EPITHET]). Selected during ingestion; bracket fields pre-populated from GroundTruth data where mappings exist.
Curator	A Chronicle team member with elevated permissions who drafts, edits, and submits ChronicleEntry blurbs for approval. The final human-in-the-loop before player-visible publication.
Prime Architect	Cody — final approval authority for all Maker Chronicle design decisions, implementation specifications, and integration architecture.
PARTIAL Status	An ingestionMeta.ingestionStatus value indicating a ChronicleEntry was created with incomplete GroundTruth

Term	Definition
	enrichment due to API unavailability or batch sub-request failure. Always requires curator manual verification before publication.
God Mode System	The NexusMUD administrative backend used by makers to perform world-state actions. The authoritative source of AscendantEvents.